Collaborative Group Work with ChatGPT in Higher Education

Ira Herawati Sumarlin

Uppsala University, Department of Informatics and Media, ira.herawati-sumarlin.8749@student.uu.se

The increasing presence of generative AI in higher education has led to significant research into how students use these tools. Still, far less attention has been given to generative AI use in collaborative contexts, despite the importance of group work and project-based assignments in many higher education systems. This thesis examines how student groups working on a collaborative design task integrate ChatGPT into their shared reasoning. Based on a qualitative study of six groups at universities in Uppsala, Sweden, the analysis develops three themes that together describe collaborative engagement with generative AI: ChatGPT as a bridge that brings new knowledge and alternative perspectives into their shared workspace, group reasoning unfolds as a social practice, and groups actively maintain their epistemic agency through distributed verification and iterative questioning. The findings show that collaborating with generative AI in a group setting is far more complex than using it individually. ChatGPT becomes part of a group's collaborative reasoning process, where the results are discussed, evaluated, and negotiated, and where contributions actively shape the direction the group pursues, rather than simply serving as a resource for information retrieval. It emerges as a social and reflective practice that student groups actively build together in real time. As a result, the interaction becomes collaborative rather than simply a tool for obtaining answers. The thesis provides an empirically grounded account of this work, with practical implications for how higher education can support the use of collaborative AI as generative AI becomes integrated into academic life.

CCS CONCEPTS • Human-centered computing • Human Computer Interaction (HCI) • Empirical studies in HCI

Additional Keywords and Phrases: collaborative task, group work, generative AI, ChatGPT, human-AI collaboration, higher education

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1 INTRODUCTION

The use of artificial intelligence (AI) in education is inevitable since it has already become part of everyday study routines [8, 22]. Generative AI, such as ChatGPT, AI systems that generate responses through prompts, can support information retrieval, explanations, and writing [10, 12]. Surveys covering 109 countries reveal that students regularly use generative AI in their daily study activities, including writing, learning concepts, and understanding course materials [7, 9, 27]. Understanding how students actually use it in practice is crucial for institutions to respond and balance its potential, maximizing benefits while minimizing challenges [8, 12].

Recent research has largely focused on the adoption of generative AI and on how intervention design can help students develop new skills while addressing concerns such as over-reliance on AI, uncertainty, and bias [8, 10, 12, 20]. Generative

AI shows strong potential for individual learners, particularly because it can personalize learning according to each student's needs and characteristics [12, 20]. However, much less is known about how students use generative AI when working collaboratively in groups, and about the role the AI plays in these interactions. As the landscape of knowledge collaboration evolves, educational settings increasingly recognize generative AI as a sociotechnical mediator that opens new dimensions of collaboration between students and knowledge [28, 34]. From this perspective, collaboration goes beyond simply dividing tasks; it involves coordinating perspectives, negotiating meaning, and developing a shared understanding of a common problem. When generative AI is part of these interactions, its output can become the object of discussion, evaluation, and refinement. Therefore, understanding how this happens is not only an educational question, but also an HCI question concerning how technology mediates collaborative work and shapes collective reasoning. This shift is important because collaborative learning plays a crucial role in fostering students' accountability for their own success and encouraging active participation in the learning process [2]. As Stahl [2006] suggests, learning happens through social interactions in which meaning is negotiated through dialogue, the use of tools, and the gradual development of shared understanding [35].

Although interest in using generative AI as a partner in education is growing, its role in practical group interactions has received limited attention. Francis et al. [8] suggest that project-based learning may help address these challenges, as this model requires students to engage in a process over time, from the initial research phase through the design process and task completion. In this context, when generative AI becomes part of these social learning processes, concerns about its opacity highlight the need for explainable, human-centered systems in educational settings [2, 8, 22]. This, in turn, highlights the importance of encouraging students to interrogate generative AI systems to maintain human agency and responsibility in academia, rather than passively accepting answers without understanding the sources and explanations [22].

This thesis addresses these issues by exploring students' use of generative AI during routine tasks in higher education. The aim is to contribute empirical and conceptual insights into how generative AI is used in group study and what these insights imply for responsible, agentic human-AI collaboration in higher education. Therefore, the study is guided by research questions as follows:

RQ1: How does generative AI function within the distributed cognitive system of student groups during collaborative design tasks?

RQ2: In what ways does AI output shape the direction of group reasoning during collaborative tasks?

By answering these questions, this thesis intends to contribute to Human–Computer Interaction and Computer-Supported Collaborative Learning (CSCL) research with a grounded explanation of how generative AI becomes integrated into student group work and how human–AI interaction shapes the trajectory of ideas in higher education.

2 BACKGROUND

2.1 Generative AI in Higher Education

Generative AI, which enables interactive dialogue through chat interfaces, has become increasingly relevant in higher education and has been studied for over a decade [12, 22, 26]. Large language models (LLMs) such as ChatGPT and Claude offer several benefits, including personalized learning, rapid feedback, and support with complex tasks [8, 10, 29]. Chatbots and virtual assistants have also been used for tutoring, writing assistance, and even as motivational companions, reportedly offering benefits such as greater availability, faster response times, and a friendly tone [20, 26]. As a result,

tools like ChatGPT and Claude are now widely integrated into education, transforming teaching and learning practices, particularly in computer science education [12, 29].

Surveys of higher-education students found that many use generative AI for brainstorming, summarizing texts, and finding research articles [9, 27]. These findings indicate that generative AI is already embedded in students' everyday learning routines, while also introducing new risks. Students may let generative AI lead collaborative problem-solving, which raises concerns about over-reliance on AI-generated outputs, including hallucinations or ambiguous answers presented with high confidence [13, 20, 22, 29]. Because LLM-based systems often lack explainability and are difficult to interpret into human-friendly concepts [12, 22], these concerns are difficult to address in practice.

Trust in generative AI is, therefore, a central issue because it raises questions about the accountability of AI-generated outputs. Unlike calculators, they rely on probabilistic models, rendering their decision-making processes less transparent and often characterized as “black boxes” [22]. Miller [23] further argues that transparency for such systems cannot be achieved in a short time of use but must be sustained over time through displaced transparency via training, explanation, planning, and debriefing to calibrate users’ trust and align their mental models with those of the system. Such explanations enable mental model reconciliation, synchronizing human and AI understanding and calibrating trust through lifecycle processes [23]. Yet dismissing AI entirely is not a viable option, given its demonstrated efficiency and effectiveness as a professional tool [12, 38]. Effective use, therefore, requires active engagement, including questioning and critical interrogation of the system. As interrogation is fundamental to human communication, McDermott et al. [22] argue that it should also be central to the design and evaluation of human-AI interactions.

Students, therefore, need not only conventional skills but also the ability to solve problems by strategically using AI technologies across different contexts, including creating AI-driven solutions to benefit society [12, 25]. Institutions can support this by promoting, redesigning assessment, and setting clear rules [8, 12]. However, this support has mostly focused on individual learners, leaving open the question of how students negotiate AI use in collaborative group settings, which this thesis addresses. Incorporating generative AI into current assessments allows students to develop both skill sets simultaneously [12]. This is crucial for curriculum preparing graduates to use generative AI responsibly, as AI systems take on increasingly active roles in collaborative work, making it important for students to maintain agency and critical engagement. In practice, however, this balance is not always achieved, and collaboration can become passive acceptance or simple task delegation.

However, student–generative AI collaboration should not be understood as a simple task delegation. Effective collaboration requires students to position AI as a cognitive partner rather than a replacement for human thinking [13, 22]. This involves negotiating meaning, critically evaluating outputs, refining prompts, and integrating AI-generated suggestions into their own reasoning processes [8, 22]. When approached in this way, collaboration becomes an iterative dialogue in which responsibility remains with the student [13]. Designing educational practices that support this balanced interaction is important to ensure that AI extends and supports human cognitive processes, rather than diminishes human agency and collaborative learning [2, 12, 20]. Understanding what this process looks like in practice, how cognitive work is distributed between students and AI requires a theoretical framework for analyzing distributed cognitive activity, which the following section addresses.

2.2 Distributed Cognition

Distributed Cognition (DCog) theory offers a perspective on cognition beyond the individual mind and treats cognitive processes as distributed across people, artifacts, and time rather than solely within the mind. Unlike the traditional view, which treats cognitive processes as isolated within individual minds, DCog sees knowledge as extending beyond individual

minds into external environments such as tools, physical and social environments, and technologies like AI [16, 30]. It moves the unit of analysis from an individual mind to a socio-technical system [16]. From this perspective, there is no gap between cognitive processes and the external world, cognitive processes are extended into and rely on external representations, making them elements of a cognitive system [16]. Hollan et al. [16] describe DCog as a new foundation for human-computer interaction research, where the unit of analysis is delimited by functional relationships among elements in the system rather than by the boundary of a person. They identified three main forms of distribution: first, cognitive processes distributed across members of a social group, where social organization serves as a cognitive architecture that supports interaction and idea-building. The second form is coordination between internal and external structures, in which artifacts and the environment are actively used to externalize and transform thought. Finally, distribution occurs over time, with the product of earlier events shaping the nature of subsequent cognitive processes [16]. Beyond seeking individual minds, it also focuses on actions related to cognitive processes. To study "in the wild," DCog incorporates methods such as observation, video, and interviews to follow how actions, artifacts, and representations can support memory reasoning and aid decision-making in real settings [16].

Building on this foundation, recent work in human-AI cooperation has adopted DCog to explore how cognition is distributed, including within AI systems [33]. To understand these interactions, a deep understanding of how human teams communicate with the support of technology, such as AI, is required [33]. Riedl et al. [30] reconceptualize AI not as a neutral tool, not as passive assistant, but as an actor that reshapes the social and cognitive fabric of teamwork, showing how AI-generated language influences linguistic alignment, shared attention, shared mental models, and group identity in human-AI teams. They describe AI as a “social forcefield” that reorganizes the distributed cognitive architecture of teams by seeding vocabulary, redirecting focus, and leaving durable linguistic and cognitive traces even after the AI is no longer present [30]. Like other cognitive frameworks, distributed cognition encompasses memory, decision-making, reasoning, and learning.

In this thesis, DCog provides a framework for analyzing student-generative AI collaboration as a distributed cognitive system comprising students, generative AI such as ChatGPT, and digital artifacts such as chat logs. Drawing on Hutchins, as cited in Hollan et al. [16], this analysis examines how cognitive work is distributed across group members, between internal reasoning and external representations, such as prompts and AI outputs, and over time, as prior AI-generated text and earlier decisions shape later interactions. Building on Riedl et al. [30] and Guo et al. [13], the study explores how generative AI acts as a collaborative partner, influencing linguistic alignment, shared attention, and idea development in student groups, and how its role affects the configuration of distributed cognition during group work.

2.3 Postphenomenology

Postphenomenology, developed by Don Ihde, provides a deeper way to analyze technology by treating it as actively shaping how people relate to the world [19, 32, 36]. In hermeneutic relations, the technology presents a readable rendering of the world that the user interprets to understand the world beyond it [19]. In an alterity relation, the technology is encountered not as a window onto the world but as a quasi-other, something whose claims can be questioned and whose authority must be examined rather than assumed [19]. The two types of relationships are not completely separate. Users can move between them or use both depending on the task they are doing. This movement between relationships reflects what Ihde [2009] calls multistability, the principle that the same technology can have different types of human-technology relationships depending on the context, use, and perspective of the user. Rather than simply occupying one fixed position in human experience, technologies tend to be multistable; that is, they can emerge and function differently depending on what the user brings to the interaction and what the situation demands [19].

While postphenomenology has historically focused on relations between users and individual technological artifacts, recent contributions have argued for extending the framework to consider technologies as parts of larger socio-technical systems and collective practices [5, 21]. The recent study is closer to the artifact-focused end of this spectrum, examining ChatGPT as a specific conversational technology that nonetheless operates within wider ecologies of academic practice. Hongladarom and van der Vaeren [18] argue that ChatGPT goes beyond the kinds of mediating technologies Ihde originally analyzed because it does not merely transmit information about a pre-existing world but generates an interpretive layer of its own, shaping the form in which knowledge appears to users.

Combining these perspectives allows us to examine both how cognitive work is shared and how technology shapes that process [31, 36]. A postphenomenological perspective helps to understand how generative AI mediates that system by highlighting some possibilities and shaping what counts as a useful contribution. In this sense, the AI system is not merely a passive instrument, but a mediating presence that participates in configuring the group's collaborative activity [19]. It makes it possible to trace not only what students and generative AI produce, but also how the generative AI outputs help shape the direction of discussion, the visibility of certain options, and the students' sense of what is possible or appropriate to do next. In other words, DCog helps explain where cognitive work happens, while postphenomenology helps explain how technological mediation shapes the form that work takes.

3 METHODOLOGY

To gain a comprehensive understanding of how students collaborate with generative AI in group settings, this study employed a qualitative approach that combines task-based observation and interviews. Users were divided into groups in collaborative sessions, followed by individual post-task semi-structured interviews.

3.1 Participant Selection

Participation was limited to students currently enrolled in higher education. Additional requirements included familiarity with and prior experience in group work, as well as in using generative AI for individual study. To reduce sampling bias, participants were recruited from multiple student organizations and academic programs across Uppsala University and SLU (Sveriges Lantbruksuniversitet). Recruitment was conducted via direct messages and WhatsApp groups. Snowball sampling was also employed, where participants were encouraged to recommend peers who met the study criteria. This strategy ensures a broader range of perspectives, while still targeting students familiar with group work and generative AI. They were informed about the purpose of the study, participation requirements, and the expected duration. The study involved 12 participants across six disciplinary groups, consistent with the purposive sampling approach adopted. This sample size aligns with empirical guidelines for achieving saturation in focused qualitative interaction studies (9–17 participants) [15].

Table 1: Participant Profiles

Participant	Group	Gender, Age	Field of Study	Education Level	University
1	1	Female, 29	Life Science	Master's degree	SLU
2	1	Female. 24	Life Science	Master's degree	SLU
3	2	Female, 28	Health Science	Master's degree	Uppsala University
4	2	Female, 32	Health Science	Master's degree	Uppsala University
5	3	Female, 28	Computing and Linguistic	Master's degree	Uppsala University
6	3	Male, 29	Computing and Linguistic	Master's degree	Uppsala University
7	4	Female, 27	Life Science	Master's degree	Uppsala University
8	4	Female, 24	Life Science	Master's degree	Uppsala University
9	5	Female, 25	Design and Computing	Master's degree	Uppsala University
10	5	Male, 40	Design and Computing	Master's degree	Uppsala University
11	6	Male, 31	Engineering	Master's degree	Uppsala University
12	6	Male, 33	Engineering	Master's degree	Uppsala University

3.2 Data Collection

In the first stage of data collection, each group of two students participated in a 45-60-minute collaborative session. During the session, participants worked on a realistic, open-ended task, and used ChatGPT as a collaborative resource. The task was inspired in part by Francis et al. [8], whose recommendations highlight the value of open-ended, process-oriented tasks for fostering critical engagement with generative AI. The project involved developing a Sustainable Health Monitoring Device for Sub-Saharan Africa (Appendix A.1). Each group contributed based on their programs and expertise, such as designing a power supply suited to Sub-Saharan African conditions, determining what is available, and assessing what happens when it runs out. They also designed a prototype feature to address the lack of digital literacy, limited screen familiarity, and cognitive load experienced by users, as well as how they report this by working across the linguistic diversity of Sub-Saharan Africa. No framing manipulation was introduced, and students were encouraged to discuss with their group and use ChatGPT naturally. Each group could vary the intensity of its use based on the group's agreement.

ChatGPT was chosen as it is currently the most widely used generative AI tool among students, 606 out of 1,265 university students reported using it for academic tasks, far exceeding alternatives like Copilot (67) or Gemini (51) [24]. All interactions with ChatGPT were collected through exported chat logs that documented the entire conversation during the task. To capture the broader collaboration process, including group discussion, the session audio was also recorded via Zoom Meeting.

After completing the task, the interviews were also conducted and audio-recorded via Zoom. Each participant took part in an individual 15-20 minute semi-structured interview exploring their experiences and perceptions of ChatGPT's contribution to the task, how they distributed cognitive work with ChatGPT, and their decision-making processes during collaboration (Appendix A.2). Following Galletta [2013], the interview questions were open-ended to allow participants to narrate their experiences, while retaining flexibility within the interview process and maintaining a close alignment with the research objectives [11]. This approach ensures that interviews complement observational data, providing insights into cognitive strategies, reasoning, and functional roles in ChatGPT.

3.3 Data Analysis

The data were analyzed using reflexive thematic analysis (RTA). Braun and Clarke [4] describe RTA as a distinct approach to thematic analysis that emphasizes researcher reflexivity, theoretical engagement, and active interpretation throughout the analytic process. Furthermore, this method recognizes the researcher's active role in interpreting the data, allowing analytical insights and theoretical perspectives to shape the development of themes [4]. The analysis used an abductive approach, moving iteratively between theoretical concepts and patterns emerging from the data. RTA allows findings to be grounded in the theoretical framework while remaining responsive to the research context and insights emerging from the data [4]. The abductive approach guides the analysis, moving back and forth between theory and collected data [6].

The analysis process using RTA still follows the six-phase workflow from Braun and Clarke [3], but is more flexible, incorporating reflexive processes, theoretical engagement, and active interpretation to identify themes [4]. In this study, the process began with familiarizing oneself with data from four sources: observations of group behavior during the collaborative task, group conversations recorded during the session, participant prompts submitted to ChatGPT, shared artifacts produced, and individual post-task interviews. Rather than treating these sources sequentially, the analysis draws on them in combination, allowing patterns observed in one source to be interrogated against and enriched by the others. Initial codes were generated from the data as the next phase, focusing on recurring actions, discussions, and practices observed across the dataset. Consistent with RTA, coding was treated as an interpretive process through which meaningful patterns were identified and developed. The coding process was carried out manually based on highlights from all sources. Codes showing analytical continuity across sources and groups were developed into candidate themes, which were subsequently examined through the theoretical frameworks of DCog and Postphenomenology (Appendix A.4). The themes generated from these initial codes were reviewed to ensure they were appropriate and addressed the research questions. Reviewing themes was a crucial step in this study, ensuring that the resulting themes were not repetitive or too broad and needed to be separated [1]. This movement, from data to code to theme to theory, reflects the reflexive and iterative nature of RTA, in which the researcher's theoretical positioning shapes but does not predetermine the analysis. The themes presented in the following section are therefore grounded in empirical material and brought into conversation with theory to generate findings that extend beyond what either the data or the frameworks could produce independently.

3.4 Ethical Consideration

To protect participants' privacy and the researcher's integrity, all participants were required to sign an informed consent form (Appendix A.3). The form outlines the observation and interview procedures and the study's confidentiality measures. During the tasks and interviews, the researcher verbally requested permission to record and informed participants that they could decline to disclose any personal information they did not wish to share. All recordings were securely processed and stored on the researcher's local device. Shared materials, including chat logs and interview transcripts, were anonymized and did not contain personal identifiers. Raw data will be destroyed upon project completion.

4 RESULTS AND ANALYSIS

This chapter presents the study's main findings. Through behavioral observations and conversations among students during the task, as well as their discussions with ChatGPT, shared artifacts, and post-task interviews, an analysis of materials from all groups and students led to initial coding. Three themes were identified concerning the use of generative AI in group work.

4.1 ChatGPT as a Bridge for Knowledge and Idea Expansion

4.1.1 Filling Knowledge Gaps

Across most groups, ChatGPT first entered the discussion after 10-15 minutes, usually when team members needed information that others in the group did not have. The group engaged with ChatGPT at specific moments, especially when it reached the limits of its collective thinking at the beginning of the task. The most consistent pattern was that ChatGPT helped fill the knowledge gap by providing information. From a DCog perspective, ChatGPT served as a cognitive aid that the group turned to when their own resources were insufficient. This pattern is also observed by Sidji et al. [33], who found that in human-AI collaborative tasks, AI contributions can become part of a team's distributed cognitive system as members extend their reasoning through AI as a shared resource. This was clear in several prompts used by the groups, which were mostly direct, information-seeking questions about specific uncertainties, showing how cognitive load was redistributed across the group and ChatGPT, embodying a form of distributed cognition. By offloading exploratory work to ChatGPT, the group was able to direct more of its attention to higher-order reasoning together. This reflects what Azevedo [2005], as discussed by Atchley et al. [2], notes: offloading lower-level work to AI tools can free students to focus on higher-level thinking.

"What are the common livestock in sub-Saharan Africa and possible zoonotic diseases?" prompt-G1
"What language do sub-Saharan African communities speak?" prompt-G3
"What diseases of Sub-Saharan African people are mainly faced and are being diagnosed late?" prompt-G4

As an external resource, ChatGPT introduced more information, expanding the group's representational space [16]. In several observed cases, this made group discussions more varied by adding new ideas for consideration, as one group asked, *"What stage can the saliva-based TB diagnosis detect?"* then continued, "*What other staining that's applicable for TB saliva detection?" (Prompt-G4).* Group members also acknowledged using ChatGPT when they needed information beyond their capabilities, *"They provide a lot of information that we don't have […] and something that we don't really know" (Interview P2)*, which aligns with the idea of complementarity, which suggests that humans and AI offer different but mutually reinforcing capabilities [14]. Unlike traditional search engines, which often left students with fragmented information, the conversational nature of ChatGPT allowed them to co-construct understanding, almost like a peer dialogue. It also facilitated the exploration of alternatives because the task was not handled entirely within individuals' minds or only through peer discussion, instead, part of the thinking process was shifted to ChatGPT [2, 33]. The group's engagement with ChatGPT began by filling knowledge gaps, continued by sparking new ideas, and then shifted to helping spark new discussions, as will be explained further in the next sub-theme. As the group continued to engage with ChatGPT, their confidence grew noticeably, not because they became passively dependent on it, but because it helped them articulate what they did not yet know.

4.1.2 Introducing Alternative Perspectives and Possibilities

Once integrated, the group's cognitive network expanded to include ChatGPT. Beyond filling factual gaps, some groups worked with ChatGPT to expand perspectives or frameworks they had not previously considered, as one student described: "*At first, it's more about the knowledge, and after that, we do more of brainstorming, like, is it possible to do this? Is it possible to do that?" (Interview P1).* They entered ChatGPT several times because they seemed to want to break out from their own perspective. ChatGPT also helped propose options when team members disagreed, rather than just relying on peer debate. Members positioned ChatGPT as an intermediary between two sides that were sometimes difficult to decide

between. This suggests that the group engaged with ChatGPT in an orienting mode: they took the possibly relevant options from ChatGPT but also offered a comparative analysis of the suggestions [17].

Interview P6: "It would be a neutral middle ground to confirm whether whose assumptions were correct."
Interview P3: "I think ChatGPT is the third person, yes, and provides the third opinion and acts as a middleman when we have different opinions."

Analytically, the significance lies not simply in the use of ChatGPT, but in the group's shared recognition of its own limits and their joint choice to externalize cognition through it. Reaching out to ChatGPT becomes a shared group process rather than an individual action. At a functional level, ChatGPT began shifting from a passive tool to an active participant, assigning simple tasks like option generation to lighten human cognitive load [17]. Observation showed this happening organically, suggesting that ChatGPT has potential not only to extend group reasoning but also to reshape its dynamics.

Building on the patterns above, ChatGPT functions in most groups as an extension of the existing cognitive system. It is used when group members lack the relevant knowledge, when they need to expand the range of options, or when cognitive load needs to be distributed. The group's reasoning was distributed across both peers and ChatGPT, with each contributing in their own way to the same functional system. Peers contributed shared knowledge, prior experience, and ongoing interpretation, while ChatGPT extended the group's capacity by providing previously unavailable information and introducing possibilities the group had not yet considered. In this arrangement, ChatGPT became an external representation resource that participated in the ways information moved and transformed within the group's collective reasoning, rather than simply a tool that passively assisted.

Following Ihde [19], the group's engagement aligns with a hermeneutic relation, in which the group reads the world and its own emerging understanding through ChatGPT's textual rendering. At the same time, this access is never neutral. Every technological mediation emphasizes certain aspects of experience while reducing others [19]. Therefore, the way ChatGPT presents answers can influence how the group's knowledge and possibilities are discovered, not just based on what the group asks.

4.2 Group Reasoning as Social Practice with ChatGPT

Beyond bridging knowledge gaps, observations revealed a social coordination mechanism operating during group collaboration with ChatGPT. Several patterns were observed across groups, such as beginning the discussion by sharing perceptions, how they generated ideas and negotiated, and how they shared the results of their agreement.

4.2.1 Joint Problem Framing

The entire group began the discussion by sharing each team member's perspective, building a shared understanding from which they could then engage with the task together. Observations showed that they opened the conversation by sharing their prior knowledge, constructing a shared interpretation of the task, and identifying what they already knew and what was considered relevant. This approach makes peer communication the foundation of group collaboration before ChatGPT enters the process.

"I want to start discussing first, like, what I mean. How should we approach that?" Conversation-P6G3
"Before the design solutions […] we should think about one design means context in exists" Conversation-P10G5

During this process, the group also often negotiated what to ask ChatGPT, aligning on a common direction and clarifying the information they wanted before using ChatGPT's help. One student raised the question explicitly: *"What are we trying to ask? […] should we ask, like, is it possible to build a device?" (Conversation-P7G4),* and what another group said, *"Maybe we can ask ChatGPT for the Q fever, what is the most likely, the best sample from the animal." (Conversation-P1G1).* This stage can be understood as "pre-thinking", similar to what Guo et al. [13] explored in their research, where students brainstorm ideas through peer discussions rather than relying directly on ChatGPT. By conducting peer discussion first, the group activates its collective cognitive resources, builds a common framing of the problem, and creates evaluation criteria before ChatGPT input enters the system. Beyond reducing over-reliance on ChatGPT through a "pre-thinking" process, this peer discussion also develops communication skills, as students need to understand individual perspectives by articulating them collaboratively.

Observations revealed that some groups revised their prompts collaboratively, adjusting the wording of their questions before sending them to ChatGPT. One student described this process: *"We want to kind of like concise the wording so we understand it more […] The other person suggests changing the wording, and sometimes we structure the questions together before typing into AI." (Interview-P4)*. These prompts functioned as mediation artifacts that clarified ChatGPT's outcomes and aligned them with the group's intended meaning. From a postphenomenological perspective, the prompt is not an expression of what the group wants to know but a textual object that already shapes what ChatGPT can render, aspects of the group's framing that enter the prompt will be emphasized in the response, while aspects that remain implicit will be reduced [19]. The group recognized that miscommunication in the prompt could shift the direction of the discussion that followed, which is why constructing the prompt was treated as an important collective task rather than left to individual work.

This pre-thinking work aligns with what Sidji et al. [33] describe in their study of human-AI cooperation, that effective collaborative reasoning depends on the shared understanding team members build with one another, and that the introduction of AI into a team complicates this process by introducing a hybrid contribution that the team must reason about together. In the observation data, the joint problem framing that came before each prompt can be read as the group's effort to strengthen this shared understanding before exposing it to ChatGPT's contribution, a sequence that gives the group a solid foundation for receiving and evaluating ChatGPT's output.

> *Interview-P8: "We ran through our specific tasks in our specific study, then we directly focused on addressing the problem through our perspective […] after that, we actually accumulated all of our questions in mind."*
> *Interview-P9: "ChatGPT entered a little bit after our initial discussion. So, we read, and there was a small discussion. We talked a bit out loud, and then we gave the context to it."*

Read together, these perspectives suggest that AI is embedded into group discussions as part of an ongoing social practice [16], not as a substitute for human-to-human reasoning. The shared mental model built through peer discussion is therefore not merely foundational but the basis for collaborative engagement with ChatGPT.

4.2.2 Output as Facilitators of Evolving Ideas

Following the discussion of aligning perspectives and negotiating questions, another visible pattern was the group's engagement with ChatGPT's output once it appeared. When ChatGPT provided an answer, the group most often read it together, and this collective reading transformed the output from individual retrieval into a group-level cognitive process. Each member participated in responding, providing feedback, whether by challenging or building on it.

Interview P3: "The idea comes from what we have learned in our study […] I think there's some particular point at which the idea developed, it's just like adding up, every time we discuss."
Interview P4: "We read it together and discuss the answer given […], but ChatGPT also suggested some improvement from the idea that we gave. So, I think it plays a part in the development of the suggestion."

In this setup, ChatGPT functions not as a final answer but as a trigger for the evolution of group ideas, helping the group combine or transform what is already in their minds with what ChatGPT contributes [33]. The stimulating effect of reading together took several forms. At times, the output encouraged the group to reflect on their thought processes and draw additional insights from what they had just read. In Group 3, for example, one member responded to ChatGPT's output by recognizing a new dimension of their problem.

"Oh, nice, interesting. I didn't know this. […] This would be our challenge." Conversation-P6G3

At other times, the output produced did not replace the group's reasoning but instead opened new questions. While they may address the group's initial questions, they also generate further lines of inquiry that the group chose to pursue through follow-up prompts or continued peer discussion. These moments often overlapped, but distinguishing them shows that this process was not a single step. It was a series of generative effects made possible by the group's collective reading.

ChatGPT's textual output, as a new external representation entering the group's shared space, illustrates what Hollan et al. [16] describe as the propagation of distributed cognition, a process in which representations are coordinated and transformed across people and artifacts, with cognitive work occurring within these transformations. The output reshaped the group's collective representational state, and its stimulating effect depended on how the group made sense of it through collective reading and interpretation. The relation here is hermeneutic in Ihde's sense, but this access is never neutral [19]. The new directions taken by the group's reasoning are therefore shaped not only by what the group brings to the reading, but also by what might emerge through technological mediation.

4.2.3 Working on Outputs Together

When ChatGPT begins to provide output and ideas, this rarely marks the end of the discussion for each group. Instead, the group continues to work with the output as a shared object. Students explain how they engage with ChatGPT answers, considering the pros and cons of the output based on their prior knowledge. In line with 4.2.1, the pre-thinking process provides a foundation for discussion. When ChatGPT's output appears, it enters the system as a new representation of state, outside the pre-thinking of the internal processes in each member's head and the external ones articulated in the conversation. A student said, *"After ChatGPT gave us the response, the list, […] we chose from that option" (Interview P8).* The process of considering the output of this ChatGPT is one of the aims, which is to place it in relation to what the group already knows, so it can be connected to their developing position. A student explained that this was done through discussion, *"We discussed the answer and whether we agreed with that, and whether we wanted to incorporate its suggestions into our ideas" (Interview P6).* The importance of this comparison process is that it is carried out together by group members, if done individually, coordination occurs only within their internal representations, not in a shared representational space. That is what makes it called cognitive distribution [16].

"It's very complicated. So, should we start searching for the disease?" Conversation-P7G4
---- "What disease of Sub-Saharan Africa people that mainly faced and being late-diagnostic?" Prompt-G4
"I think we should pick TB, because its spread is bigger, so we know the disease." Conversation-P7G4

----"Explain more about the pathogenesis of TB and the biomarkers for diagnosis" Prompt-G4
"I don't think it's difficult to detect because it's easy to spread, right?" Conversation-P7G4

A moment captured in the conversation above with group 4, where they received general information and were then given an idea by ChatGPT. Based on this idea, they appear to have made a collective decision through discussion. The clause "*we know the disease*" indicates that they already had prior knowledge of the disease but obtained the idea for the task at hand from ChatGPT. The choice is re-emphasized by the word "*right?*", indicating they already understood it and wanted to incorporate ChatGPT's answer.

This second theme demonstrates a shift in how students and ChatGPT engage with one another during collaborative work. Where the first theme showed ChatGPT functioning as an extension of the cognitive system, the second theme reveals that engagement with ChatGPT is itself a social practice. The first theme's request-driven outputs feed directly into the second theme's discussion phase, showing how the initial gap-filling becomes the foundation for the development of ideas mediated by communication. Because the cognitive system extends to all peers, ChatGPT, and the artifacts they co-produce, such as prompts and outputs, the system operating in the second theme no longer sees ChatGPT as an external resource but as a coordinating extension [16]. The key feature of this theme is its recursive nature: discussing to reach agreement on the direction of completing the task, the group creates prompts, the questions the group asks to shape the result, the result reshapes the group's reasoning, and the reshaped reasoning shapes subsequent questions. A postphenomenological reading suggests that this integration also has an interpretive texture. The group's relationship with ChatGPT operates primarily in a hermeneutic relationship [19]. At the same time, however, the group positions ChatGPT as a quasi-other, moving towards a social role with its own presence, which it has not yet fully achieved.

4.3 Maintaining Epistemic Agency in Collaborative Group Work with ChatGPT

While the previous theme demonstrated how groups relate to ChatGPT as a social practice, this third theme complements it by showing how the output and social practice are maintained through their own reasoning. Across the six groups, they not only received answers from ChatGPT to develop pre-thinking processes, whether for general information or to expand ideas, but also actively worked to maintain their own judgment at times. For instance, they verified by asking other sources, reformulating prompts to clarify the answers, and determining how much to rely on information from ChatGPT. These activities demonstrate that epistemic agency, their authority and capacity to evaluate, accept, or reject knowledge, is actively carried out through distributed work.

The work of maintaining epistemic agency can be understood by examining the concept of interrogability, as proposed by McDermott et al. [22], in the context of AI-driven education. To determine the veracity of information presented by AI and the reliability of its source, this interrogation capability is necessary [22]. The difference from the questioning process is important because the interrogation process requires more focused, critical questioning [22]. This theme explains how students maintain epistemic agency with interrogability. Starting with the verification process, then how they try to reformulate their communication to get the answers they need, and the limits of ChatGPT's answers.

4.3.1 Verification as Distributed Work

In addition to integrating ChatGPT's responses to develop ideas, the group varied in how they responded to its output. The first activity seen was when several groups tried to critically compare the output with what they already knew. They then questioned the source of ChatGPT's answers, either by asking ChatGPT directly for the source or by checking sources

outside the platform. This verification was not an afterthought but rather part of the group discussion activity. Unlike subtheme 4.2.3, which focused on using ChatGPT's answers appropriately, this subtheme assessed the trustworthiness of the output for discussions. The process in this subtheme is rarely separate from the integrative process in the second theme, as participants often work on the output together. In practice, these two modes are not entirely separate but can appear within a single interaction, depending on the output. When ChatGPT's output raises doubts about what the group already knows, the group verifies its reliability.

Verification took several visible forms, one of which was comparing ChatGPT's output with the group's own existing knowledge. At one point, a student evaluating ChatGPT's claims about bacterial behavior drew on their own understanding to argue: *"I think bacteria [...] So there's no bacteria being shed into saliva or like a biofilm." (Conversation-P8G4).* The phrase "*I think*" here draws on students' prior knowledge as an active resource for interrogation material, which must be compared with ChatGPT's claims. At other times, they tried to verify by consulting other sources they considered necessary, such as references or citations. A student said*, "I usually ask the source of the information […] If I want to use the information from the ChatGPT, I will ask for the real source." (Interview P1)* and reinforced by statements from other students, *"Sometimes I can be skeptical about their response […] I don't always rely on ChatGPT. […] I also check other sources, whether this is true or not." (Interview P5).* Another form of verification observed was students checking and reassigning questions to produce more specific qualities, a process aligned with the lines of discussion in the second theme. One student described this process, *"We already send it, but then we try to revise it […] we just like copied the first prompt and then revise it afterward […]* S*ometimes, if we ask, like, the wrong question, it will come up with a whole different explanation that we don't really need." (Interview P8).* Cross-checking of these three activities does not occur in isolation, but it often occurs together in a single exchange, depending on what the group feels needs verification.

Group verification practices help maintain reliability in systems where cognition is shared among people and tools. Hollan et al. [16] argue that in such systems, no single source has the final say. Information is valid not because an authoritative person says so, but because it has been verified through multiple channels, such as comparisons across different people, artifacts, and external references, and these channels are consistent with each other [16]. In this observational data, the group did exactly what was observed. ChatGPT was one of several sources during the discussion. Before its contribution could be incorporated into the group's reasoning, it had to align with what the students already knew and with the external references.

In the second theme, the group's engagement with ChatGPT operates within what Ihde [19] calls a hermeneutic relationship. In this mode, ChatGPT functions as a kind of translator, transforming silent digital information into a language the group can interpret and use [19]. When the verification process happens, this relationship shifts. The group no longer reads the world through ChatGPT but instead confronts ChatGPT itself, asking whether what the AI offers is trustworthy. This shift moves the group toward alterity relations, where ChatGPT is encountered as a quasi-other, a distinct entity with its own presence whose claims can be addressed and challenged [19]. This shift also enacts what McDermott et al. [22] describe as interrogability in practice, in which the group's verification work is an interrogative engagement with the AI, systematically testing the trustworthiness of the information it provides. In this observation, maintaining epistemic agency is not a passive stance but rather a coordinated set of practices distributed among members and across multiple sources of knowledge. The group does not simply accept or reject ChatGPT, rather, it builds upon it through continuous verification.

4.3.2 Holding the Line through Iterative Questioning

While the previous subtheme described how groups verify ChatGPT contributions to assess trustworthiness, this subtheme describes how groups decide whether to forward them. During the task, several activities were captured in the groups as

they considered the position of ChatGPT output, whether to accept it, when to limit it, and when not to use it. Therefore, this subtheme is not concerned with checking reliability but rather with where usage boundaries should be set and under whose authority.

Students demonstrated that holding the lines was not an instantaneous act of rejection. There was even a difference between what they described during interviews and what they did when completing the group task. In their interviews, several students described their disagreement with the ChatGPT results as if it were a seamless process: *"There was one particular part where I disagreed with ChatGPT, in which it said that the system wouldn't have to be very expensive." (Interview P6)*. Another student also explained, "*We deny that ChatGPT suggested another part […] because we have our reference for a suitable part that we can use" (Interview P11).* The phrases "*I disagreed*" and "*We deny*" indicate decisive, completed actions, as if nothing else were being done at that moment, or they explicitly reject ChatGPT's answer. Yet, when compared with observations at the same moment from recordings of group discussions and written prompts, a different picture emerges. The experiences they attempted to recall during the interviews were, in fact, the conclusions of extended iterative work, not entirely decisive and complete at that moment. The disagreement was resolved through discussion and further tested by submitting questions to ChatGPT. This does not mean their descriptions of their perceptions and experiences during the task are incorrect, they did experience disagreements, but the process was iterative.

The contrast is evident in the example of asking about costs in the reference above. During the interview, P6 explained the disagreement with ChatGPT as a single moment of rejection. However, in their group discussion (group 3), the rejection did not arrive as a final decision but rather developed as a question to be worked through. Rather than dismissing ChatGPT's claim outright, the group returned to ChatGPT with a sharpened follow-up question: *"Are these expensive? Which one is the most possible and not too expensive to be implemented?" (Prompt-G3).* This process illustrates that the group asked because the act of asking, of forcing AI to specify, compare, and contextualize, is how the group works out where it stands, in line with what was discussed in 4.2.1. Therefore, the explanation of the disagreement conveyed during the interview is not a beginning but an outcome, following the group's explanation of its interaction with ChatGPT, which involved asking several questions to map what ChatGPT can and cannot contribute to this issue. Allowing ChatGPT to get inside each other's heads and keep asking for detailed explanations can help address transparency issues in generative AI systems.

This pattern is also evident in some of the prompts they wrote. Because the assigned task was to design a system, the group's assessment of the need for ChatGPT's contribution shifted from requests to more specific, decision-oriented ones. One group, working on a feature they considered foundational, prompted ChatGPT with: "*Can you give me the specific roles of […], because I think it is central, it can be used as […]" (Prompt-G2).* From the prompt, it's clear that the group was not just asking, but also conveying what they already thought, using the phrase "*I think it is central*" to specify their intended use. It positions the group as an authority that has already made some decisions, and ChatGPT as a contributor who elaborates on those decisions. In this specific moment, iterative questioning is not subordinate inquiry. It's a directed inquiry, in which the group's position is already partially formed, and ChatGPT is invited to contribute on the group's terms.

At several points, it can be said that the process of making decisions regarding ChatGPT's contribution to the group, through disagreement, quiet acknowledgment of limits, and the use of follow-up prompts that are more in-depth than the previous question, is a form of iterative engagement, not a pre-decision. The group builds this process by engaging in conversation with ChatGPT by repeatedly asking, evaluating, and adjusting until the position feels clear enough to act on.

Iterative questioning is a form of social organization in which thinking unfolds across people, tools, and time. Hollan et al. [16] describe cognition as the continuous transmission and transformation of information as it moves through a sociotechnical system. It involves both our internal resources, such as memory, and external structures, such as tools. Each

prompt the group submits illustrates a process of representation moving from the group's discussion to ChatGPT's response, which is then examined, reshaped, and sent back as a new question, and the cycle continues. Thinking does not happen in any single place but in the movement between them. Reaching the point where the group can say "*we deny"* is not the end of their thinking. It shows that the group has developed a clear enough understanding to decide and act on it. This process also aligns with the concept of Mental Model Reconciliation by Miller [23], drawing on Chakraboti et al. [2017], which is the process by which humans and AI synchronize their understanding of a problem when perceived discrepancies are detected, and it continues until these discrepancies are understood and repaired.

What makes this iterative collaboration feel natural is how generative AI responds to a question. Ihde [19] describes alterity relations, moments when humans engage with technology as if it were a quasi-other. Because ChatGPT can respond in everyday language, the group then behaves much like one would when talking to a partner whose answers can be countered, refined, or rejected. The alterity relation here is not a one-off interaction but an ongoing exchange, and it is in this back-and-forth that the group works out where it stands.

Overall, the two subthemes presented here illustrate how the groups in this study maintained epistemic agency in their collaborative work with ChatGPT. While these two practices address different questions, they still share the same underlying logic. In both cases, the groups not only passively received ChatGPT's contributions but also processed them. They tested them through various forms of validation and investigated them through continuous inquiry until a position was reached. Based on a holistic reading of the six groups, epistemic agency in generative AI-assisted collaboration is neither automatic nor singular. This achievement is not spontaneous, but rather distributed through the careful and intentional ways in which the groups embedded their contributions within their own reasoning.

This epistemic agency is itself a form of distributed cognitive work. Verification and iterative questioning are not the activity of a single member, but rather a coordinated effort distributed across the group. In DCog terms, this forms a propagation of representational states through the cognitive system, not merely receiving information from ChatGPT but transforming, evaluating, and calibrating it as it moves through the group.

The change in mode of relation seen in this theme can be seen in how groups shift from reading the world through ChatGPT to confronting ChatGPT itself as a quasi-other. What comes into play is not just their practice but also their understanding of what ChatGPT is, what types it is, the authority of its claims, and the kind of knowledge it generates. The group's perception changes and shifts in meaning regarding ChatGPT's epistemic status emerge through engagement. This movement is what Ihde calls multistability: the same technology can support different types of human-technology relationships depending on the needs of the situation [19]. What this study adds is that this multistability is negotiated collectively, not individually. It is done collectively through discussion to determine what kind of relationship they have with ChatGPT at any given moment. While the second theme describes how the groups integrated ChatGPT into their shared reasoning, this theme describes how they maintained the authority that kept that integration their own.

5 DISCUSSION

This study aims to contribute to our understanding of how student groups work with generative AI in collaborative tasks. As noted in the introduction, while generative AI has been widely discussed in higher education, its use in collaborative contexts remains limited. This gap is significant, given that group work and project-based assignments are central to and often form part of the assessments of many courses in higher education in Sweden, where the present study was conducted.

The discussion that follows is organized into two parts. The first part returns to the research questions and traces the findings, and the second part turns to the practical implications of these findings.

5.1 Result Discussion

The first research question of this thesis sets out to answer how generative AI is integrated into the distributed cognitive system of student groups. The results and analysis of the first and second themes offer a layered answer to this question. The first theme explains the representational side of integration. ChatGPT contributes to group discussions by filling knowledge gaps or introducing alternative perspectives that the group had not yet considered. In this case, ChatGPT extends what DCog describes as representational space, the set of resources and perspectives available for collective reasoning [30]. The second theme explains how ChatGPT contributions are integrated into the group's collaborative dialogue. Not simply used, but ChatGPT output is woven into the dialogue practice at key moments across every stage. Taken together, these two themes demonstrate that integration is an ongoing process of coordination within the group around ChatGPT's contributions. ChatGPT contributes resources, but these resources only enter the group's collective reasoning through the social practices that surround them. Sidji et al.'s [33] experiment on cooperative games showed that team cognition in such a setting cannot be reduced to the internal thoughts of its individual members, rather, it lies in the coordination among humans, artifacts, and AI. The findings of this study support that view in a different setting, collaborative design tasks in higher education.

Regarding the second research question, how the output of generative AI can shape the direction of group reasoning during collaborative tasks, the second and third themes in the analysis complement each other. The second theme demonstrates that ChatGPT output does not function as the final answer for the entire group, but rather as a starting point. They open new space for inquiry and spark discussion by frequently raising possibilities that have not been discussed in the group, thereby changing previously unexpected perspectives. In this case, the output from ChatGPT was generative rather than closed, as it opened space for broader discussion. The output influenced the direction of group reasoning by expanding the scope of discussion when the group considered it necessary. The third theme, however, shows that this broadening of discussion was not aimless, and this shaping was not something the group passively accepted. After allowing ChatGPT's output to open new directions, the groups carefully decided which directions they would accept and under what conditions. Two practices carried out this work. The first was verification as a distributed activity (Subtheme 4.3.1), in which participants interrogated the output by comparing it with their own knowledge and other external sources before deciding which contributions to adopt. The second was iterative questioning (Subtheme 4.3.2), through which groups actively guided the discussion rather than simply following where the output led, focusing on what they considered central.

Collectively, these two sides give a full answer to RQ2. The direction that group reasoning eventually took was therefore not produced solely by ChatGPT, but by the interaction between ChatGPT's suggestion and the group's epistemic agency in responding to it. ChatGPT appeared to provide direction, but it was still the group that decided how to use that influence. The direction reasoning finally takes is the product of this two-way interaction rather than of the output alone. This pattern became visible because the study drew on more than one type of data, which complemented each other. What participants described in interviews, such as how they decided to use ChatGPT, considered its answers, verified them, and at times decided to disagree, could be confirmed and nuanced by what they actually talked about in the group and what they asked ChatGPT.

This shifting interaction between the group and ChatGPT also deepens the answer to both research questions when viewed through the lens of postphenomenological theory. The first and second themes align with what Ihde [19] calls hermeneutic relations, the meaning of which is explained in section 2.3. ChatGPT's output functions as a mediator between

the group and what it represents, and the group's meaning-making process is directed at that representation. However, the third theme indicates a shift towards what Ihde [19] calls alterity relations, a mode in which technology is encountered as quasi-other. Verification and repeated questioning are no longer ways of reading the world through ChatGPT, they are ways of confronting the ChatGPT output itself, examining what it has to offer, and deciding whether and how to incorporate it into the group's reasoning. These three themes together suggest that the group's engagement with generative AI is not fixed in a single relationship but shifts between them, depending on the task's phase and the type of work the group is undertaking. This is consistent with what Sidji et al. [33] describe as the difficulty of forming stable mental models when AI contributions are integrated with human reasoning, as the relationship shifts with the work, and the group must recalibrate over time. Seen this way, the shift between relations is not a separate theoretical observation but part of the answer to both research questions.

The findings presented here demonstrate three specific ways in which the use of generative AI in collaborative tasks adds nuances distinct from individual use. First, working with ChatGPT in groups is not simply the shared use of an individual. It relies on social practices such as shared problem framings, shared reading of the output, and collaborative work to discuss what ChatGPT produces in relation to the task at hand. Second, groups develop a distributed form of epistemic agency. Verification work takes place across multiple ways among members, rather than within a single mind, and iterative questions are shaped by group discussion. Third, ChatGPT's influence on group reasoning is iterative and mediated rather than direct or fixed. ChatGPT serves as a starting point, but the group performs the calibration of its contribution. These features are often missing in individual use and are clear when used in groups.

Building on these findings, the collaborative framework adopted here does not claim that ChatGPT is inherently a collaborator, but better captures its role in these interactions as observed in this study. The perspective of using tools to obtain information, similar to how people use sophisticated information sources, and of cognitive artifacts as material objects that carry representational structure and participate in distributed cognitive work, explains their supporting function. However, they do not fully capture how its contributions become part of the group's collective discussion, evaluation, and knowledge construction. Overall, collaboration in ChatGPT-supported group work is a dynamic process in which human reasoning and AI contributions are collectively engaged with, evaluated, and integrated during task completion.

5.2 Practical Implication

The findings presented in the previous section partially answer the broader question of how higher education institutions and educators can support students who are increasingly expected to incorporate generative AI into their academic practices. During the interviews, several points emerged about their feelings about using ChatGPT in group discussions. While students noted several benefits of using ChatGPT in their group work, such as faster shared understanding and more efficient discussion, they have also been uncomfortable using it in front of their peers, even though it does not actually break any rules. One student said, *"In the university environment, it's a bit taboo for us to open ChatGPT when working with others,"* and another said, "*The rules haven't been fully established,"* especially regarding their use in group work.

The groups in this study used ChatGPT because the assignment explicitly permitted it, and the activity took place in an observable environment with clearly outlined norms of use. However, outside of these conditions, participants said that the use of AI in group work tended to remain hidden or avoided. Students cannot develop this use independently, especially as they themselves say it is "*not completely set yet*". Therefore, to support norm formation, institutions need to develop guidelines. Some institutions, including Uppsala University and SLU, where this study took place, have already established guidelines and support structures for educators regarding the use of generative AI in teaching and assessment [37, 39]. Francis et al. [2025] argue the importance of a comprehensive generative AI literacy program for educators and students.

In this case, institutions cannot expect students to collaborate with AI if educators do not fully understand how assessment design and practice will be.

The findings of this study indicate that the presence of ChatGPT in group work not only adds a new tool to the collaborative environment but also changes the nature of the work itself. With ChatGPT as part of the group process, students' attention is distributed differently: alongside their discussion, they must also evaluate ChatGPT's contributions, assessing their reliability, source, and relevance to the task. While ChatGPT offers benefits such as improved group work effectiveness, it also introduces new types of uncertainty that need to be discussed together. In this sense, collaborative project work with generative AI is phenomenologically distinct from group work without it, in terms of the experience it offers.

The next concern is how this applies to group project work. For some courses that allow this form of collaboration, one suggestion for course designers is to create collaborative assignments that explicitly invite generative AI into group projects, with the condition that it be treated as a contributor whose use is anticipated rather than prohibited. Tasks like rapid prototyping, multiple solutions, and collaborative comparison of alternative drafts can make projects more exploratory and design-oriented. Atchley et al. [2] make a similar argument for higher education more broadly, advocating that AI be included as a secondary collaborator in group assignments and that students be required to make their use of AI visible and reflective. To improve project assessment, students could be required to submit an AI usage log explaining how they used AI output and what decisions they based on it. The group could later be assessed based on how they present the project results, including their reflections on collaborating with AI. The findings of this thesis support and extend these arguments with empirical grounding. The groups in this study did not use ChatGPT to replace their own thinking, they used it to expand their thinking, while repeatedly drawing on their own knowledge to verify, refine, and sometimes reject what ChatGPT offered. Assignments that explicitly invite this collaboration are unlikely to impose unfamiliar practices on student groups. But the hope is that they can improve their communication skills, foster critical thinking, and maintain agency in decision-making.

A further implication concerns what students need to learn to use AI effectively in group settings. The students in this study were all familiar with using generative AI individually as a prerequisite before participating. However, despite this individual familiarity, collaborative AI use was new to almost all of them. The practices they used to achieve this are not the same skills developed through individual use of AI. These are skills that need to be developed by groups, not individuals, and are shaped through coordination rather than through the competence of a single person. This distinction is crucial for how AI literacy is taught. AI literacy should be treated as a social competence, with group-structured opportunities to negotiate norms for acceptable AI use, evaluate the reliability of outputs, and decide how to attribute contributions, which can help preserve the social and dialogic nature of collaborative learning with generative AI. Although Uppsala University and SLU have established guidelines for AI use in individual learning and assessment, the collaborative dimension of AI literacy may remain largely unmentioned. Sidji et al. [33] observed a related challenge in their study of human-AI collaboration, when AI contributions are mixed with human reasoning in a team, group members must develop the ability to reason about the hybrid contributions, applying what they call an AI filter to assess how much of an idea comes from the AI and how much from human teammates. The groups in this study demonstrated parallel competencies in their verification practices and iterative questioning. The implication is that AI literacy in higher education needs to address this collaborative dimension. Curricula could include it as an explicit element of AI literacy, for example, by making it visible in courses involving group projects, where collaborative engagement with AI is most likely to occur.

5.3 Limitation and Future Research

The study draws on a relatively small sample of 12 participants, distributed across 6 groups, all enrolled at universities in Uppsala, Sweden. While this scope was appropriate for the depth of qualitative analysis the thesis required, it also means the findings reflect a specific institutional and cultural setting. Collaborative AI use in other higher education contexts, with different academic traditions or institutional policies, may unfold differently. The task examined was a collaborative design task, group work on other kinds of academic tasks, for example, problem-solving in technical domains, literature-based research, or written assessment, may engage AI in ways that this study has not captured. Therefore, the open-ended task design, the academic profile of the participants, and the encouragement of natural AI use described in the data collection together formed conditions that may be context-sensitive. This study also focused on capturing group reasoning, rather than individual learning outcomes. The use of ChatGPT in this study may differ from other generative AI systems in response delivery, structure, and information presentation, which may impact the findings. The findings should therefore be read as describing what becomes possible under these conditions rather than as a universal feature of collaborative work with AI.

Future research arising from this thesis would focus on collaborative AI use in conditions closer to everyday academic practice. Design research in partnership with a course that includes a group project assignment, in which the use of generative AI is explicitly permitted as part of the assignment instructions. Such a study would extend the present work in several useful ways. Although it would necessarily take longer than the present work, since it requires following groups through a complete course cycle, it would produce a richness of evidence that shorter observational designs cannot reach.

6 CONCLUSION

This thesis has examined how student groups working on a collaborative design task integrate generative AI into their shared reasoning. In analyzing the six groups, three main themes emerge. First, the groups used ChatGPT to fill knowledge gaps and open new perspectives they had not previously considered. Second, they incorporated their findings into a collaborative conversation by engaging in shared framing, collective reading, and joint efforts on the outcomes. Lastly, they actively maintained their epistemic agency by continuously verifying and questioning. Thus, collaborating with generative AI in groups differs significantly from using it individually. It is a social and reflective practice that is evolving in real time among student groups, highlighting the need for higher education to provide thoughtful support as generative AI increasingly becomes part of collaborative academic work.

AI USAGE STATEMENT

In this thesis, participants used ChatGPT during observation sessions as part of the research. The interaction between participants and ChatGPT was the primary source, and the results were collected as data artifacts for analysis. Other AI tools were also used for reference searches, language refinement, and transcription. Perplexity was used to help identify references to recent studies and theoretical frameworks and to clarify key concepts. Grammarly, integrated with Microsoft Word, was used to check sentences, correct spelling, and improve grammar during the writing process. The generation of initial transcription of the recorded interviews and group conversations was assisted by Otter AI, without uploading personal data to the AI. This process was also continued manually for more accurate results. Apart from the AI-assisted parts described above, the remaining parts of this thesis, including the content, analysis, and conclusions, are my own. All AI suggestions were critically reviewed before being integrated.

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A. APPENDICES

A.1 Collaborative Task Design

Title: Sustainable Health Monitoring Device for Sub-Saharan Africa

About This Project

You are invited to join a design team developing a Sustainable Health Monitoring Device (SHMD) for deployment in sub-Saharan Africa. The device is intended to support community health workers (CHWs), local health providers who are often the only point of care in remote areas, in monitoring and responding to health conditions in the communities they serve. This project brings together specialists from different disciplines. Each group contributes its own expertise to one part of the device. Your group represents the specialists in your field. You will not work with the other groups during this session, your contribution stands independently. There is no single correct answer, no required format, and no expected output length. Work on this the way you would naturally approach a collaborative design problem in your field.

The Context

Sub-Saharan Africa faces profound challenges in healthcare delivery. Access to trained clinicians is limited in rural areas. Community health workers are the frontline response to these challenges. They work in the field, often alone, with minimal equipment and limited diagnostic capability. They may have limited formal education and varying levels of digital literacy. Any device that reaches them must work reliably in these conditions, be practical, durable, low-power, and usable without specialist training. The SHMD concept is a wearable or handheld health monitoring device that CHWs can use to assess patients in the field, transmit health data to a clinic or health system, and receive alerts or guidance in response. How exactly this works, what it measures, how it is powered, how it communicates, and how it is used are what the specialist teams are each working out from their own disciplinary angle.

Role

Your group decides how to engage with the problem from your disciplinary perspective. Discuss what your group's specialism brings to this project and decide how to engage with the problem from your disciplinary perspective. Even though you have not seen the other groups' work, briefly consider where your component overlaps with theirs.

Using ChatGPT

ChatGPT is available throughout the session. Use it however your group finds natural, there are no rules about when, how often, or in what way. Some groups may query it frequently; others may use it once or twice. Both are fine. The point is not to use it in any particular way, but to work on the problem as your group sees fit.

A.2 Interview Questions

Part 1 Task Experience and AI Use	Q1: Can you walk me through how your group approached the task today? Q2: When did ChatGPT first enter your group's work, and what was happening in your discussion just before that moment? - *Was there something your group didn't know, or something you were unsure about, that made you turn to ChatGPT at that point?* Q3: How would you describe the role ChatGPT played in your group's work today? - *Did that role stay the same throughout, or did it shift at any point?* Q4: After ChatGPT gave you a response, what would typically happen next in your group, what did you do with that answer? Q5: Were there times when your group decided not to use ChatGPT, or chose to work through something without it? What happened there?
Part 2 Cognitive Work	Q6: How did your group divide up the thinking work today? Q7: When your group decided to ask ChatGPT something, how did you figure out what to ask? Q8: Were there any moments during the task where you read ChatGPT's response and felt uncertain about it, something that didn't seem right, or that you weren't sure about? Q9: Were there any moments where one of you knew something the other didn't, or where your own expertise made you read a ChatGPT response differently? Q10: Were there any shared documents your group created or used during the task? - *(if yes) How did those help your group think together?*
Part 3 Idea Evolution and Decision-Making	Q11: Were there moments where ChatGPT seemed to lead your group's thinking in a direction you hadn't planned? - *(if yes) Can you describe specific moment when a ChatGPT response changed what your group was doing or thinking?*

	Q12: Think about the final idea or direction your group arrived at by the end of the task. If you trace it back, where did it come from? Can you identify any specific moment that shaped where you ended up?
Part 4 Reflection	Q13: Do you have any final reflection about today's task? - *Is there anything about how you or your group used AI today that you noticed but we haven't talked about yet?* Q14: How does today's experience compare to how you normally use ChatGPT, either on your own or with others?

A.3 Informed Consent

Contact: Ira (ira.herawati-sumarlin.8749@student.uu.se)

You are invited to take part in a research study exploring how small groups of students use generative AI (specifically ChatGPT) as a collaborative resource during open-ended academic tasks. Before you decide whether to participate, it is important that you understand why the research is being conducted and what it will involve. Please read this information carefully and ask questions if anything is unclear.

Purpose of the Study

This study investigates how generative AI tools are integrated into the distributed cognitive systems of student groups. Grounded in Distributed Cognition and Postphenomenology theory, the research examines how interactions among students, AI-generated outputs, and external artifacts (e.g., notes, shared screens) occurs during collaborative work.

What Participation Involves

If you agree to participate, you will be asked to:

1. Join a group of 2-3 students for a single collaborative session lasting approximately 45-60 minutes via Zoom.
2. Work with your group on a realistic, open-ended design task. You will have access to ChatGPT as a collaborative resource and are encouraged to use it naturally.
3. Take part in an individual semi-structured interview of approximately 15-20 minutes after the group session. The interview will explore your perceptions of AI's contribution, how cognitive work was distributed with the AI, and your decision-making processes.

Data Collection

The following data will be collected during and after the session:

1. Zoom audio recording of the full group session.
2. Exported ChatGPT conversation logs documenting all AI interactions during the task.
3. Audio recording of your individual post-task interview.
4. Any written or digital artifacts produced during the session (e.g., notes, diagrams, shared documents).

Confidentiality and Data Storage

1. All data will be anonymized. Pseudonyms will be assigned to all participants and groups.
2. Raw recordings and transcripts will be stored and accessible only the purposes of this research.
3. No personal identifying information will appear in any published output.

Participation in this study is entirely voluntary. You may:

Decline to participate without giving any reason and without any consequence.

1. Withdraw from the study at any time during the session without penalty.
2. Request that your data be withdrawn and deleted up to [April 2026], after which anonymization may make individual data removal impractical.

Withdrawal will not affect your academic standing, grades, or relationship with the university in any way.

Potential Risks and Benefits

This study involves minimal risk. You may feel mild discomfort discussing your AI use or academic practices. You may decline to answer any question at any time. The session is designed to mirror typical academic group work, so no unusual cognitive or emotional demands are anticipated.

There are no direct personal benefits guaranteed. However, participation contributes to understanding how AI tools function in collaborative academic contexts, which may inform future educational practice and tool design.

Use of AI Tools

ChatGPT will be used as a task resource during the session. ChatGPT is operated by OpenAI. By using this tool during the study, you acknowledge that interactions may be processed by OpenAI in accordance with their privacy policy. The researcher will export chat logs for research purposes only. You are encouraged to use ChatGPT as you naturally would, without any manipulation of how it is framed.

Consent Declaration

Please read statement below and initial the box to confirm your agreement.

O I am enrolled in higher education.

O I have read and understood the information sheet for this study and agree to participate.

Confirm your agreement to audio/video/photo recording during the collaboration task and interview

O I give my consent

O I don’t give my consent

Participant Name

_____

A.4 Result Analysis

Link to access the detailed: https://figmashort.link/ijNTAA

A.4.1 Overview of Data Sources



A.4.2 Linking Data

Conversation	Prompt	Observation Notes	Interview	Keyword	Theory
"I want to start discussing first, like, what I mean. How should we approach that?" P6G3 "how do we start? We'll start by addressing the problem first?" P8G4 "Before the design solutions, possible design solutions that we asked, what we should, what each thinks, if we should think about one design means context in exists.." P10G5		Starts with discussion and share POV	"we ran through our specific tasks in our specific study...then we directly focused on addressing the problem through our perspective...after that, we actually accumulated all of our questions in mind..." P8 "it entered a little bit after our initial discussion. So we read, was a small discussion. We talked a bit out loud, and then we gave the context to it" P9	social first, discussion first, perception alignment	DCog (extend cognition)
"Okay, so we can use ChatGPT, because it helps a lot. What was the prompt should we use?" P7G4 "What are we trying to ask? I mean, should we ask, like, is it possible to build a device, like an automated cell counter, also detection, but it incorporate fluorescence laser?" P7G4		Most of the time, they discuss and ask each other what prompts to use. Sometimes, they directly ask ChatGPT for general information out of curiosity.	"We, sometimes we type it directly, and another, the other person suggests changing the wordings, and sometimes we structure the questions together before typing into.." P4 "We always ask to each other, like, should we ask ChatGPT, or should we not? And then if we, if we agree to ask, then we just asked together.." P7 "We did discuss it first with each other.." P8	confirmation, perception alignment	DCog (extend cognition)

	G1 “what is the common livestock in sub-saharan Africa and possible zoonotic disease” G3 “what language does sub-Saharan Africa communities speak?” G4 “What disease of Sub-Saharan Africa people that mainly faced and being late-diagnostic?” G6 “how much solar radiation per year in Sahara africa”	After first discussion then start asking general information	“first, like, what can we do with this question? And after we found some difficulties, we asked ChatGPT…I guess chatGPT helps us to gain a deeper knowledge” P1 “I think it's quite helpful because they provide a lot of they provided a lot of information that we don't have, like, initial knowledge…” P2 “we didn't have that much knowledge about sustainable energy and the other fields, And we both agreed that in this way, ChatGPT was coming up with new information for the first task” P9	general information, fill knowledge	DCog (extend cognition)
“Oh, nice, interesting. I didn't know this. See, oh, see this. This is our this would be our challenge. It would need powerful hardware, which means it requires power, but the electricity is not always available there. So that's our challenge” G3 “oh yeah, this makes sense, although in rural areas, we can pick one, right? I, um, I think we should pick TB, because it's the spread is bigger, so we know the disease, co infection” G4		They read together then discuss it	"I think there's some particular point at when the idea, when the idea developed, it just like, adds up, like, every time we discuss.” P3 “after we consult the ChatGPTwe got insights and and right, we can, we can use this..” P5	developing ideas, generating new insights	Post phenomenology (hermeneutic)

	G1 “do you think it is possible to make a handheld/wearable device to detect q fever and rift valley fever infection with non-blood body fluids?” G3 “The communities will be speaking minority languages which the healthcare professionals at the hospitals that receive the data sent by the device do not speak. Whats the best solution?” G4 “is it possible to build a device that's similar to cell counter with features of fluorescence-based detection?”	The group started exploring more possibilities based on the insights and suggestions provided by ChatGPT	“At first, it's more about the knowledge, and after that, we do more of brainstorming…” P1 “But ChatGPT also suggested some improvement from the idea that we gave.. So I think it plays part into the development of the suggestion” P4 “every other prompt that we use. ChatGPT is kind of like the necessary prompt for our ideas to be developed in their reliability, afterwards we considered ChatGPT’s responses” P8	developing ideas, open more possibilities	Post phenomenology (hermeneutic)

“I mean it, it is not expensive for us, but that is also the price of the chip alone. But this is affordable for NGO.” G3 “but you can see here the limitation has low sensitivity, but, but our goal is to detect as early as possible sensitivity. But why is it low? Low in sensitivity?” G4	G2 “you said CHWs maintain registers or digital records. So what is the problem exist? cz i think it is centra, it can be used as a monitoring tool, to forecast the need of supply and even to share information between CHWs about the stock of medicine or rapid test tools”	They did not fully trust ChatGPT’s answers, so they discussed them with their partner and searched for additional information on their laptops outside of ChatGPT	“when I am still not certain with the answer, with the answer or the data provided, I will try to find it somewhere else,.” P3 “Sometimes I can be skeptical about their response, that they also like put the link of the journal that is cited from the answer, so we can check it...” P5 “there was one particular part where I disagreed with ChatGPT, in which it said that the system wouldn't have to be very expensive...” P6 “We deny that ChatGPT suggested another part, mostly for the decision specific part that we not agree with sensibility, because we have our reference for suitable part that we can use” P11	uncertainty, skepticism	Post phenomenology (quasi-other)
“You want me to share my screen? Let's write this down” P6G3 “We can make a summary. We can take notes. Okay, let's see. I'll take note, summary of our discussion” P8G4 “So let's see, have we at all. How would we frame this in our little document?” P10G5		They agreed to create a shared document to document their agreements, findings, and related notes.	“It was easier for both of us to visualize what we are writing in the same you know, we're in on the same page. And it made it easier also, for example, to split the task., we can see that each one of us progress.” P9 “I think it helped us, because when you have, if you have shared notes, it kind of forces you to work together to see what the other person sees.” P10	collaboration, coordination, sensemaking	DCog (extend cognition)

			“like in the university settings, it's a bit taboo for us to open to ChatGPT while working with others…” P3 “when I had a group discussions in university, it feels like, we avoid to use AI in a public” P7 “but it still have to be thought of how we how one would use them when we actually present something, if it was a real project. So so that in that case, it's not the rules are not completely set yet” P10	Hesitation	
			“if we are not rely on ChatGPT, we wouldn't. We still could end up here, but probably with a longer time..” P4 “I will confirm it through ChatGPT, because it is just so convenient. It saves so much time to just ask ChatGPT … then consulting the internet, the old way, where you read 1000s upon 1000s of lots of sources of internet discussions” P6 “I think especially in this first phase of collaborating with someone, when you ideas, I think that takes a lot of time in generally. So therefore, that's why we, we use the ChatGPT with the context as a first message to get that first spark you know, faster.” P9	Save time	

A.4.3 Thematic Analysis

